4.4 BASIC FOR LOOPS

COMP 2314 - Introduction to Java

Understanding Java for loops

Learning Objectives

- Define a for loop
- Identify initialization, condition, update
- Use ++ and -- operators
- Write simple counting loops

Why Use a For Loop?

- Repeat statements a specific number of times
- Useful when number of iterations is known
- Keeps code compact and readable
- Common in counting and processing data

For Loop Syntax

- for (initialExpression; conditionExpression; updateExpression) {
- // loop body
- }
- Initialization runs once; condition checked each iteration; update runs after each iteration.

Increment and Decrement Operators

- ++i means i = i + 1
- --i means i = i - 1
- Common in loop updates
- Example: for(i = 0; i < 10; ++i)

Execution Flow

1. Initialization
2. Condition check
3. Execute loop body
4. Update variable
5. Repeat until condition is false

Example: Compute an Average

- Read number of values
- Loop through each input value
- Add values to valuesSum
- Average = valuesSum / numValues

Code Walkthrough

```
for (i = 0; i < numValues; ++i) {
    currValue = scnr.nextInt();
    valuesSum += currValue;
}
```

Process each input value once